

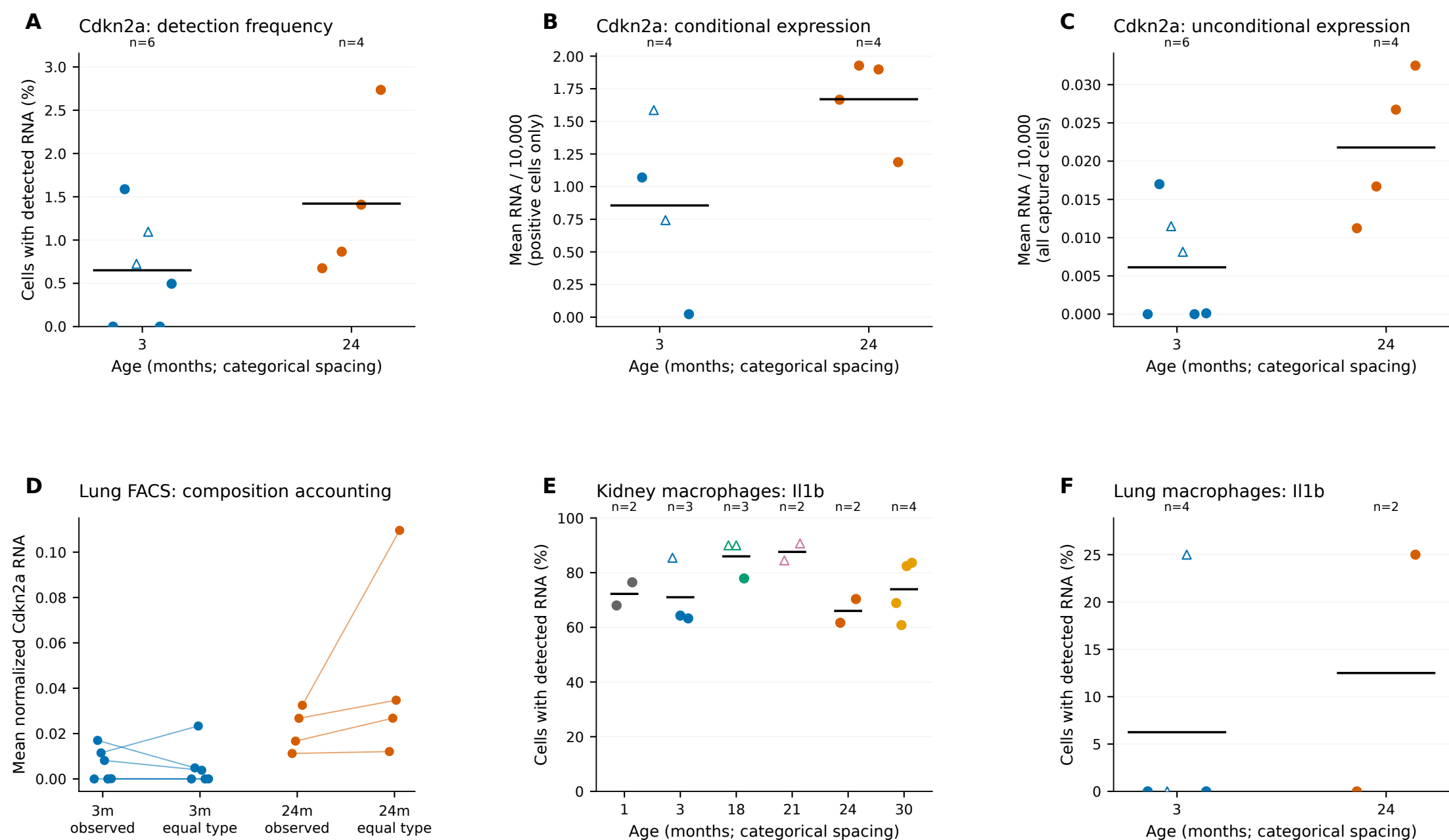


Figure 4 | Marker detection, conditional expression and tissue composition describe different endpoints.

Rationale: An increase in a tissue marker can arise from more RNA-positive cells, higher expression among positive cells or a different captured cell mixture. Separating these quantities prevents an RNA measurement from being equated with senescence or inflammatory function.

A-C, lung FACS Cdkn2a detection frequency, positive-cell mean and all-cell mean. Values use normalized RNA per 10,000; two young mice with no detected Cdkn2a have an undefined positive-cell mean and are omitted only from B. D, paired observed and equal-common-cell-type means per mouse; common FACS types cover 91.34-99.01% of captured cells. E-F, macrophage Il1b detection in kidney droplet and lung FACS. Points are mice, triangles females, circles males and black bars mouse means; n is shown.

Results: Lung FACS Cdkn2a detection averages 0.650% at 3 months and 1.421% at 24 months, while the all-cell normalized mean rises from 0.006125 to 0.021784 (n = 6/4). The panels expose animal variability, sparse macrophage support and the effect of reweighting. Reweighting is descriptive and uses incomplete common-type coverage; it is not causal mediation. Cdkn2a RNA is not a senescence assay, and Il1b RNA does not measure cytokine secretion.